

TROUT LAKE, NT, Canada

WMO: 711660

Lat: 60.440N

Lon: 121.240W

Elev: 1634

StdP: 13.85

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -40.3	(c) -34.3	(d) -48.7	(e) 0.3	(f) -40.1	(g) -41.5	(h) 0.5	(i) -33.7	(j) 18.1	(k) 17.0	(l) 16.1	(m) 19.1	(n) 2.5	(o) 120	(p) 0.720

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.5	(c) 79.0	(d) 59.2	(e) 75.5	(f) 58.3	(g) 72.5	(h) 57.0	(i) 62.1	(j) 73.0	(k) 60.5	(l) 70.7	(m) 58.9	(n) 69.1	(o) 7.6	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.1	76.8	65.6	56.3	71.7	64.3	54.5	67.1	62.8	28.7	73.2	27.5	70.6	26.4	69.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 16.7	(b) 14.5	(c) 12.9	(e) -44.8	(f) 84.8	(g) 4.2	(h) 2.6	(i) -47.9	(j) 86.6	(k) -50.3	(l) 88.1	(m) -52.7	(n) 89.5	(o) -55.8	(p) 91.4		
(4) 16.7	(5) 14.5	(6) 12.9	(7) -44.9	(8) 65.2	(9) 4.3	(10) 1.8	(11) -48.1	(12) 66.5	(13) -50.6	(14) 67.5	(15) -53.0	(16) 68.5	(17) -56.2	(18) 69.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	-7.1	2.6	8.2	30.0	42.9	54.7	58.8	54.5	45.5	31.5	10.8	-0.7		
	DBStd	25.52	15.75	13.86	15.62	11.47	8.54	6.07	5.36	6.17	7.24	8.32	12.93	15.31		
	HDD50	8788	1770	1327	1295	602	246	23	4	27	170	576	1177	1571		
	HDD65	13610	2235	1747	1760	1051	685	312	202	329	586	1040	1627	2036		
	CDD50	669	0	0	0	1	27	163	277	166	34	1	0	0		
	CDD65	14	0	0	0	0	0	2	9	3	0	0	0	0		
	CDH74	454	0	0	0	0	14	117	205	110	8	0	0	0		
	CDH80	60	0	0	0	0	1	13	26	20	0	0	0	0		
	WSAvg	5.5	4.1	4.5	5.5	5.7	5.9	5.9	6.1	6.3	6.5	6.0	5.0	4.4		
	PrecAvg	16.1	0.8	0.6	0.6	0.6	1.3	2.5	2.8	2.3	1.6	1.2	0.9	0.8		
(15) Precipitation	PrecMax	20.5	1.5	1.3	1.3	1.3	2.6	4.4	5.4	3.9	4.0	2.2	1.7	1.7		
	PrecMin	10.5	0.3	0.2	0.0	0.2	0.3	0.6	1.2	0.5	0.6	0.5	0.3	0.4		
	PrecStd	2.3	0.3	0.3	0.4	0.3	0.6	1.0	1.0	0.9	0.7	0.5	0.4	0.3		
	PrecStd	2.3	0.3	0.3	0.4	0.3	0.6	1.0	1.0	0.9	0.7	0.5	0.4	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.8	43.4	50.0	65.0	75.4	82.1	83.4	82.7	74.3	62.8	44.8	47.3		
		MCWB	31.8	34.9	37.0	47.1	53.1	55.7	62.3	62.0	57.2	49.4	37.3	37.4		
	2%	DB	30.5	37.9	43.7	57.9	69.9	77.5	78.9	76.7	67.9	53.8	39.7	37.4		
		MCWB	27.1	31.4	34.5	43.0	51.0	56.5	60.8	60.0	53.9	44.2	33.8	33.3		
	5%	DB	25.9	32.7	39.5	53.0	65.6	73.6	75.9	72.9	63.9	48.9	35.0	30.3		
		MCWB	23.4	27.9	31.6	40.2	49.3	55.3	60.0	58.2	52.1	40.9	30.8	26.9		
	10%	DB	18.8	26.9	34.3	49.0	60.6	69.9	72.8	69.0	59.8	43.9	30.4	23.7		
		MCWB	17.2	23.9	28.8	38.2	46.7	54.7	58.5	56.7	49.8	38.1	27.5	21.6		
	0.4%	WB	30.2	35.6	38.2	47.8	55.0	61.7	65.2	63.7	58.8	50.8	37.2	34.1		
		MCDB	33.4	42.5	48.5	64.7	70.0	71.8	75.3	74.8	69.7	62.6	43.2	38.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.6	31.9	35.0	43.6	52.2	59.3	62.9	61.5	56.0	45.0	33.5	29.1		
		MCDB	29.2	38.1	43.7	57.1	66.9	70.4	73.8	72.3	66.2	52.5	38.3	32.1		
	5%	WB	22.3	28.2	32.0	41.0	50.1	57.8	61.2	59.6	53.2	41.6	30.5	25.5		
		MCDB	24.6	32.8	38.9	51.6	63.6	68.8	71.8	69.5	62.2	47.9	34.4	28.8		
	10%	WB	15.9	24.1	29.1	38.7	47.8	55.9	59.8	57.4	50.5	38.8	27.1	20.4		
		MCDB	17.4	27.2	34.7	48.7	60.2	66.7	70.0	66.8	58.6	44.0	29.9	22.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.6	25.7	27.8	25.8	23.7	23.3	21.5	22.0	20.1	15.5	18.6	19.1		
		MCDBR	24.8	33.6	33.3	30.8	33.2	31.4	29.9	31.1	29.3	25.3	20.7	22.7		
		MCWBR	22.0	27.4	24.7	18.2	17.6	15.4	14.7	16.8	16.8	15.9	17.7	20.2		
		MCDBR	24.7	33.5	32.6	28.8	30.8	26.4	25.4	26.7	27.5	23.7	20.6	22.4		
	5% WB	MCWBR	22.3	27.7	24.8	17.7	17.0	14.1	13.9	15.3	16.8	15.4	17.2	20.0		
		MCWBR	22.3	27.7	24.8	17.7	17.0	14.1	13.9	15.3	16.8	15.4	17.2	20.0		
(40) Clear-Sky Solar Irradiance	taub		0.199	0.241	0.261	0.319	0.342	0.343	0.358	0.341	0.301	0.275	0.213	0.171		
	taud		2.184	2.298	2.292	2.281	2.435	2.470	2.445	2.485	2.600	2.534	2.216	2.028		
	Ebn at Noon		197	250	280	279	281	282	274	270	267	238	185	154		
	Edh at Noon		11	19	28	34	32	31	31	28	21	16	11	8		
(44) All-Sky Solar Radiation	RadAvg		135	425	984	1529	1765	1803	1699	1376	902	423	153	71		
	RadStd		9	25	54	85	117	160	113	114	96	42	21	4		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	+3.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page